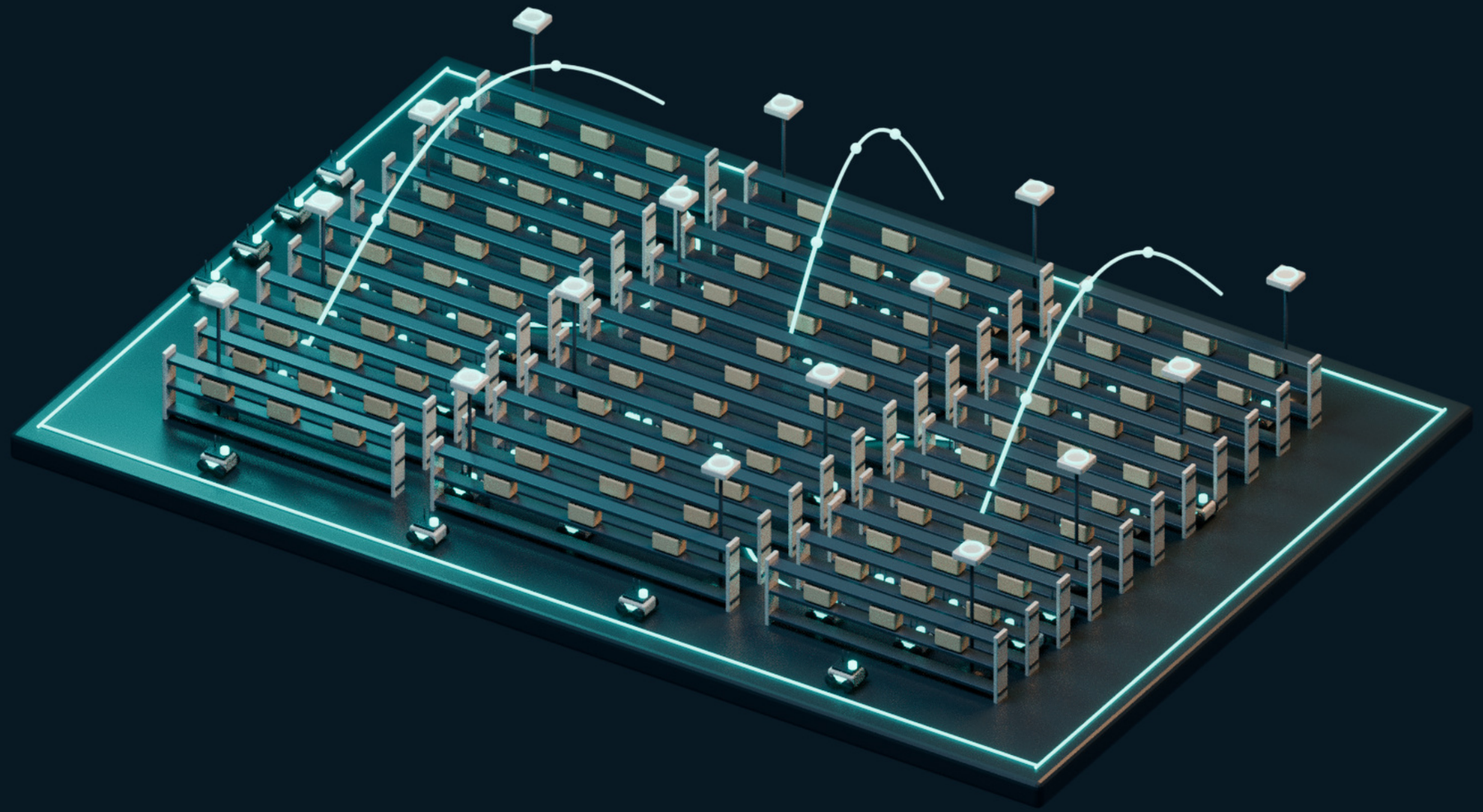
Connect the whole warehouse.

Site learning. Mission-aware decisions. Independent packet paths.

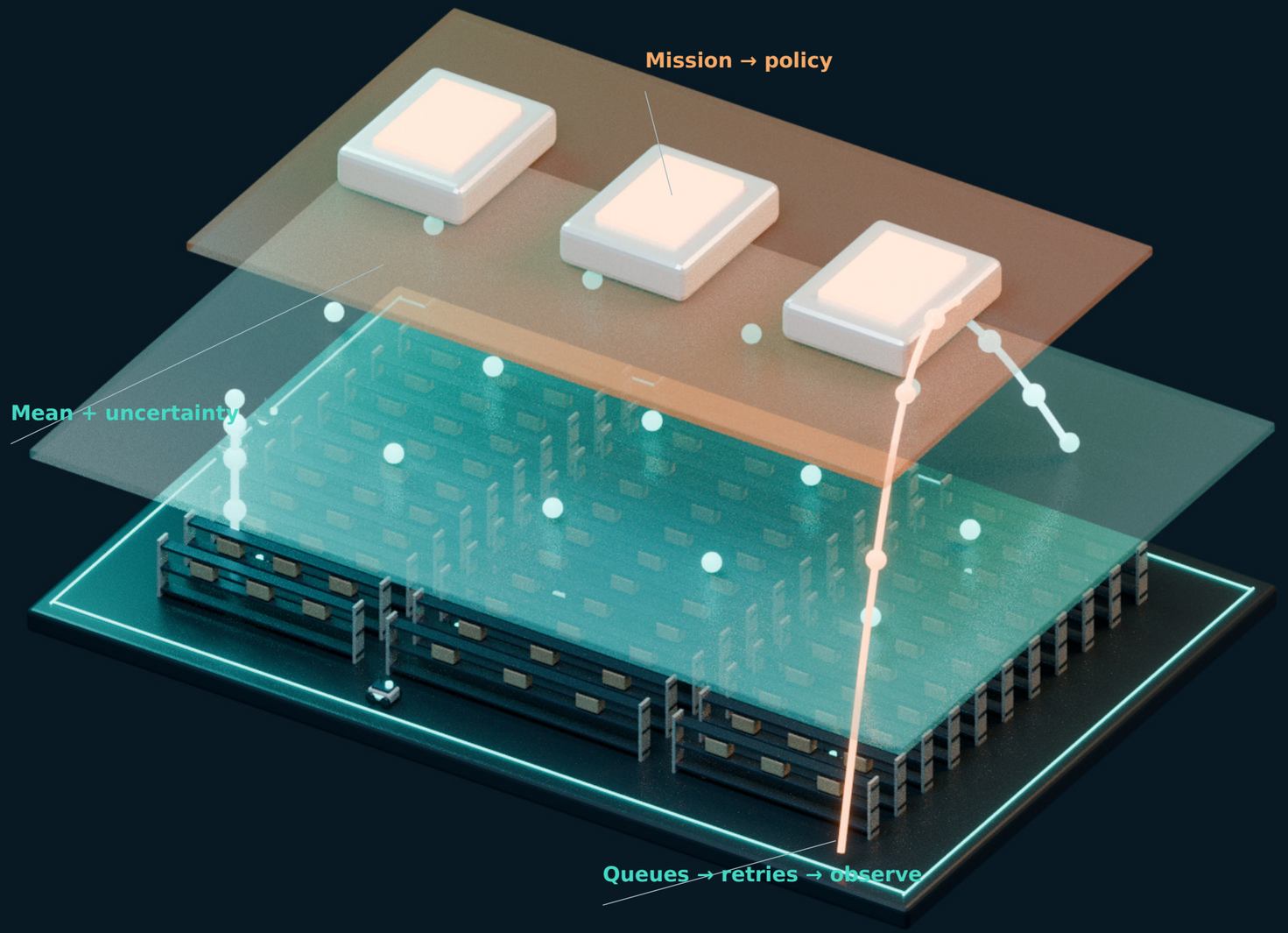


- 100 simulated robots
- 16 AP sites
- 50 ms control deadline

SIMULATION EVIDENCE / SCHEMATIC 3D MODEL

One system. Three loops.

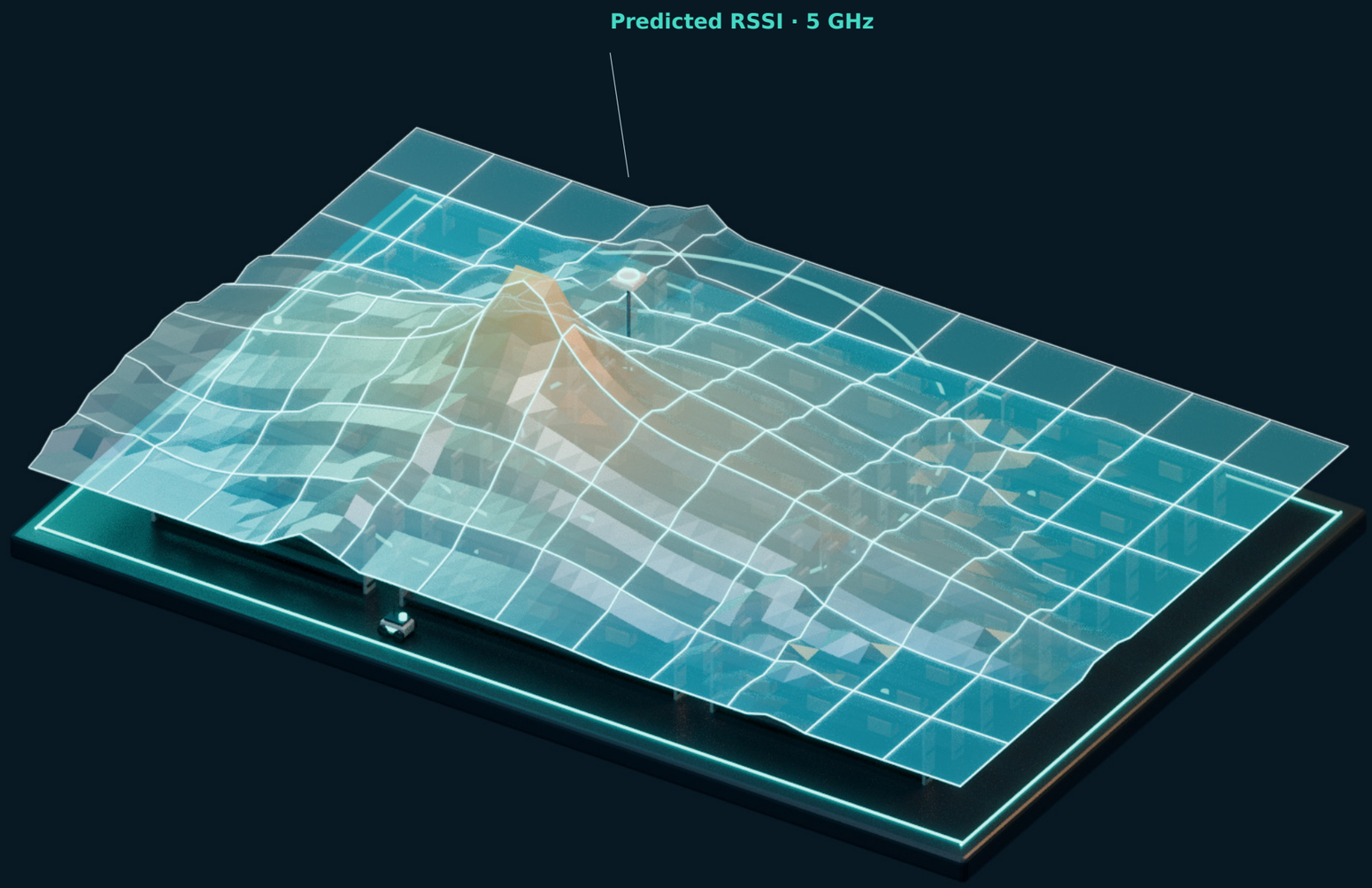
RadioMap senses. WirelessJEV decides. The executor recovers.



- Learn the site
- Plan with intent
- Recover and observe

Make radio state visible.

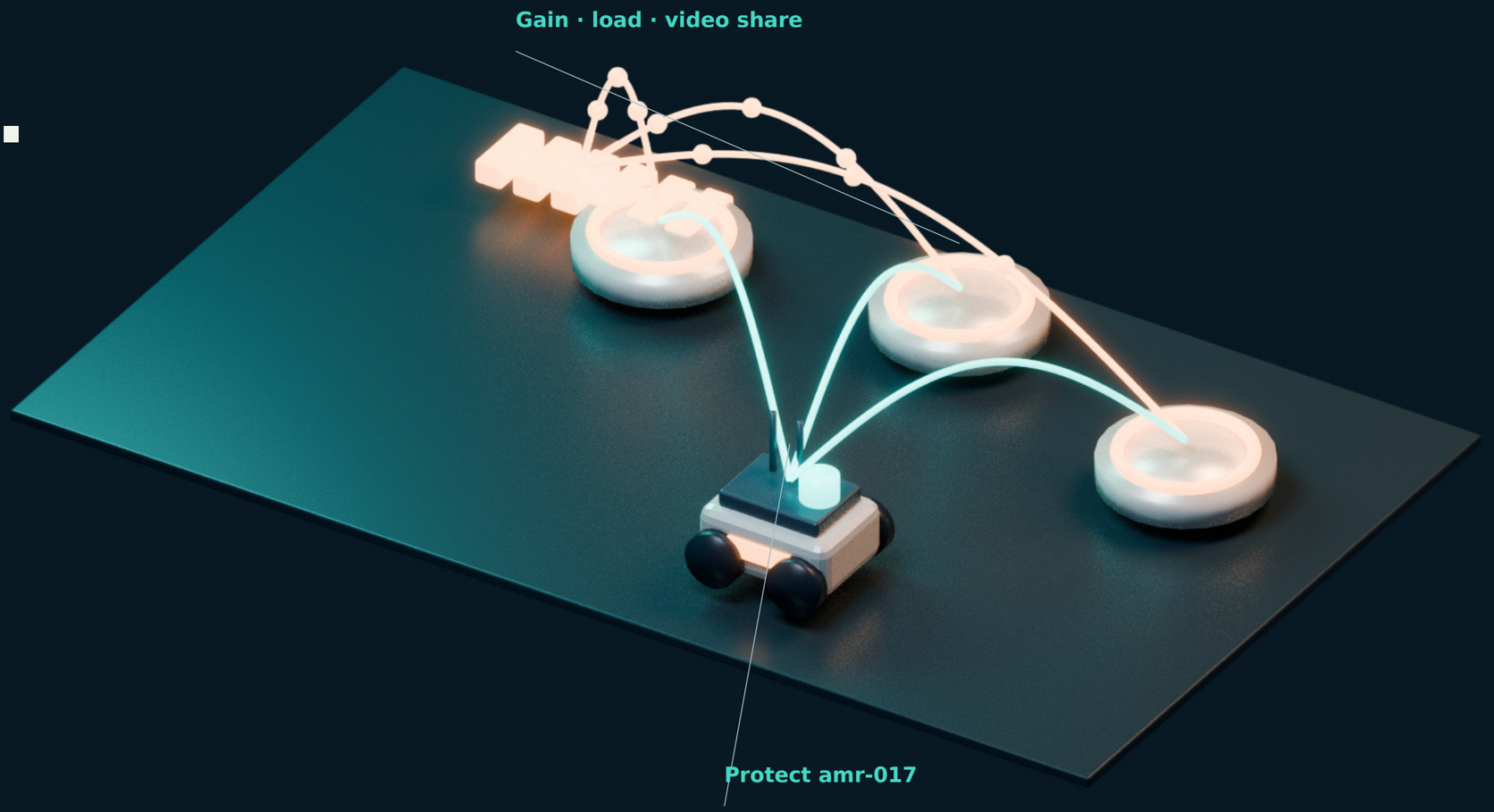
Capped scans become a site model—with uncertainty attached.



- 3.8 dB mean map RMSE
- 60-second survey
- Uncertainty still undercovers

The mission sets the policy.

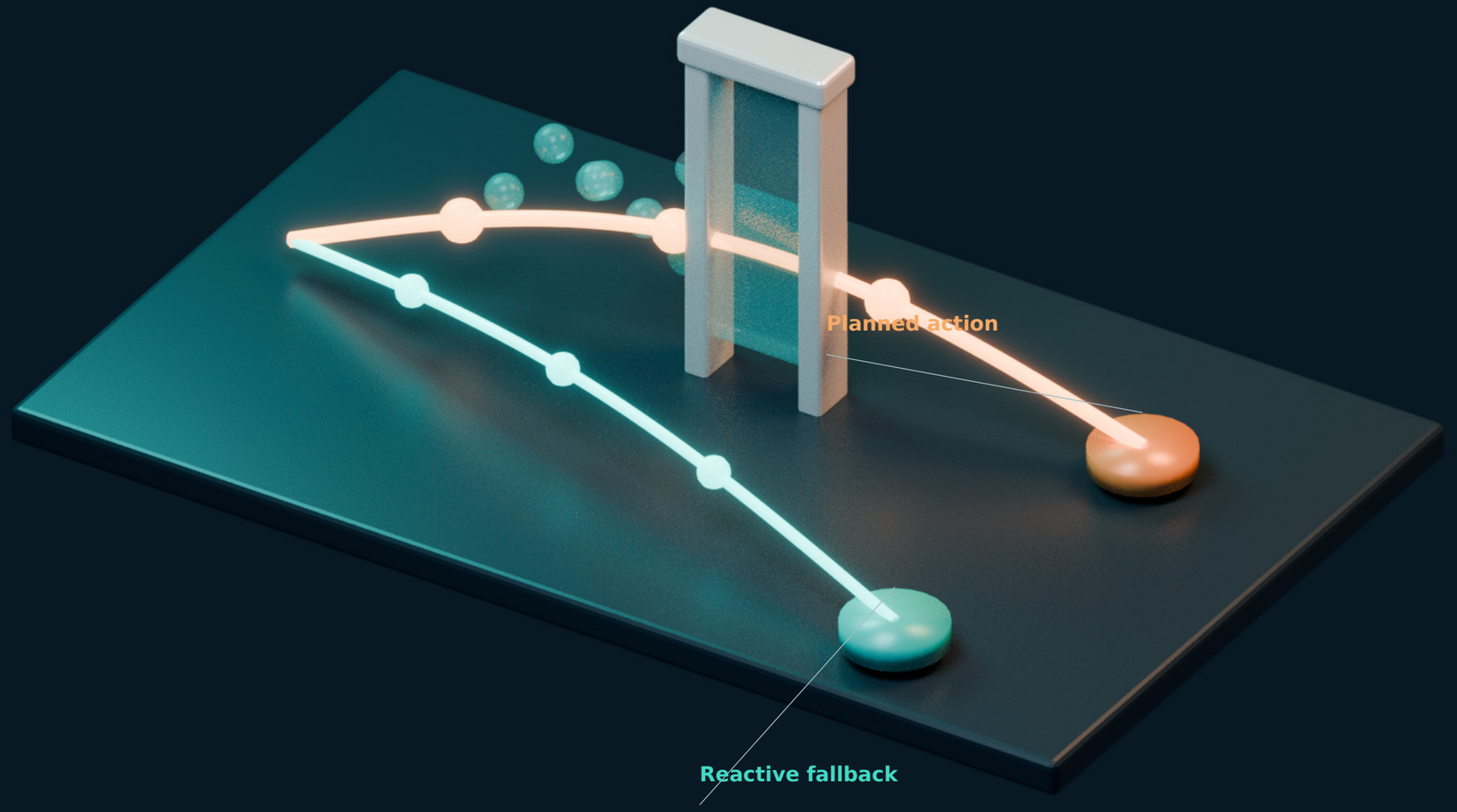
Operator priorities reach AP choice, traffic allocation and roaming.



- Named-robot priority
- Explicit policy knobs
- Language stays off the packet path

Confidence controls autonomy.

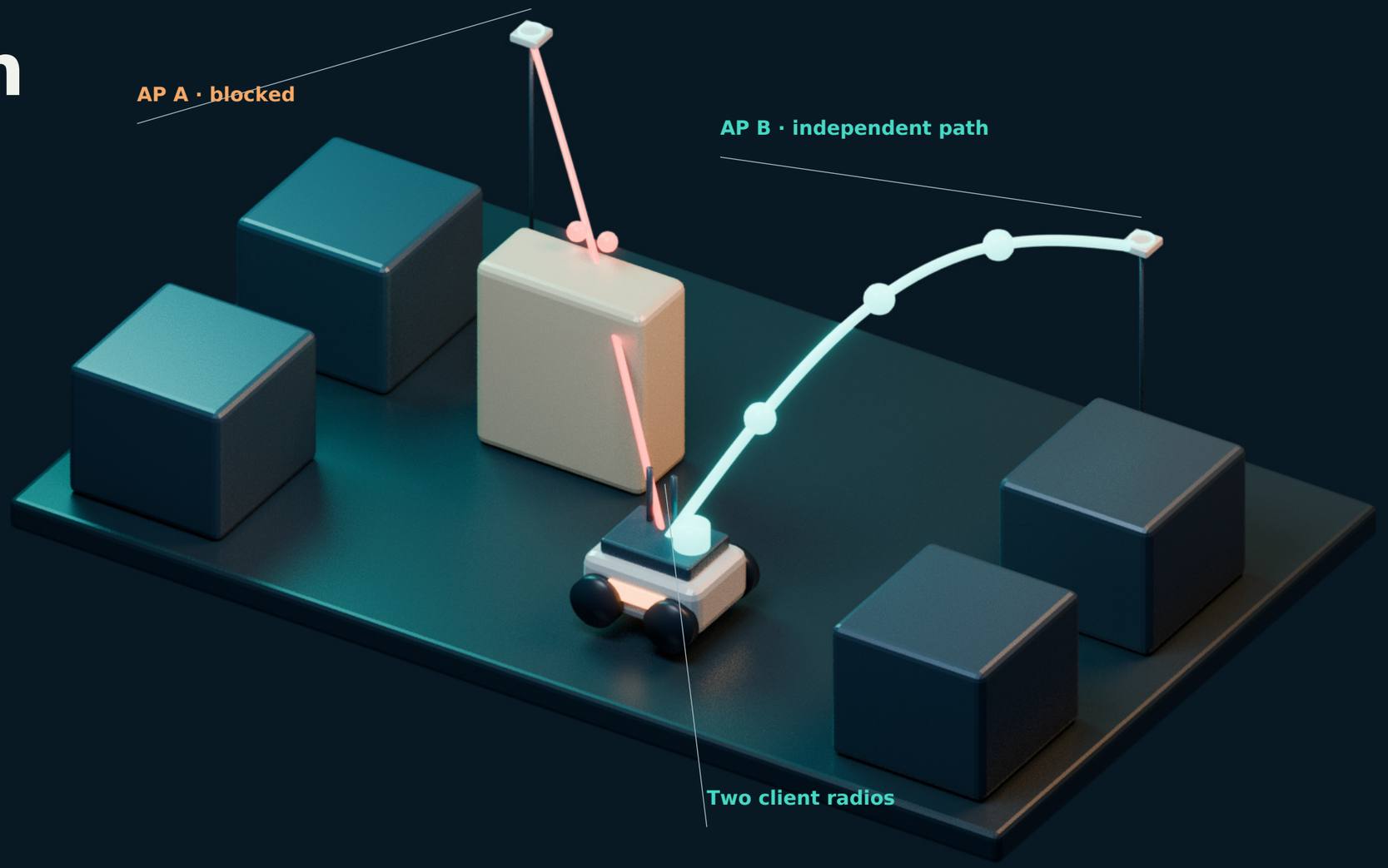
Plan where the map supports it. Fall back where it does not.



- Predicted gain / uncertainty
- Planned or reactive action
- A safeguard—not a safety certificate

Recover through a different AP.

A second geometric path can survive the first path’s blockage.

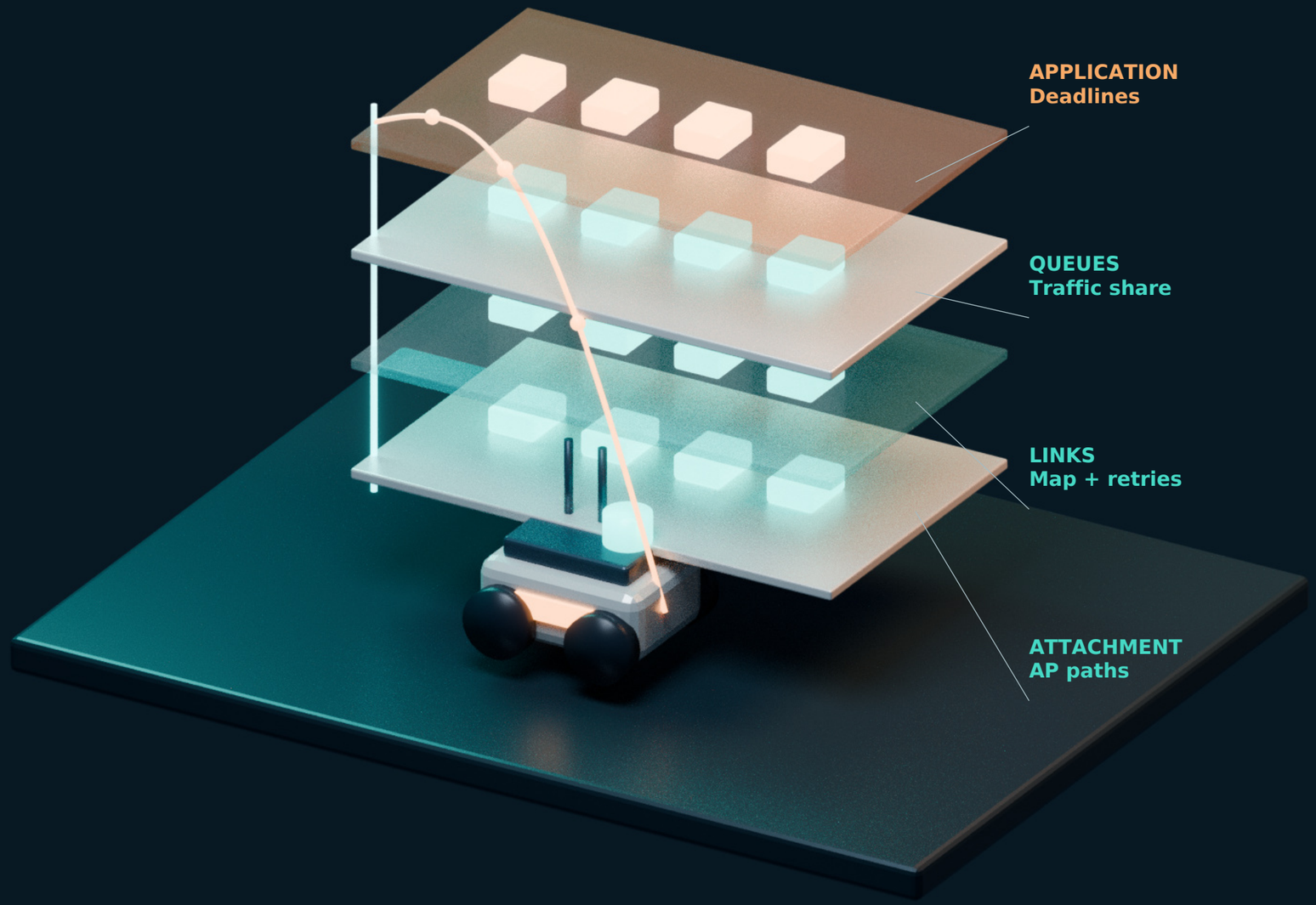


- AP A fails
- AP B remains available
- Completed result assumes two client radios

Coordinate every layer.

Deadlines, queues, link mapping and retry paths act together.

- Control: timely
- Video: on time
- Bulk: opportunistic



17× fewer missed commands.

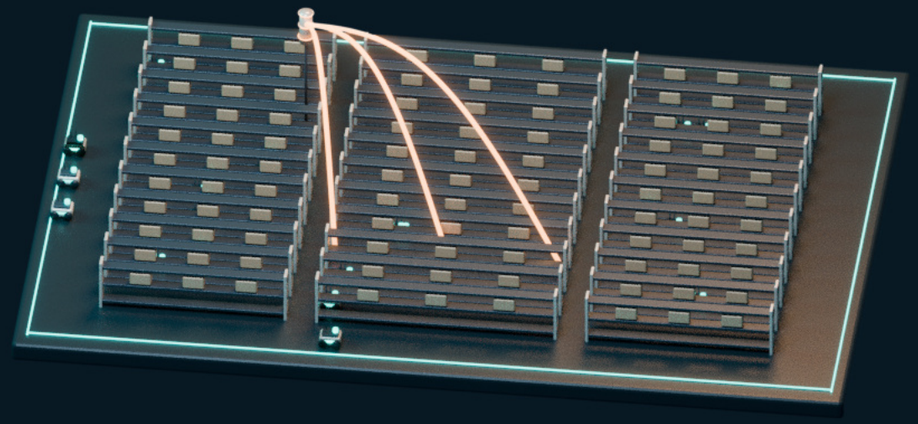
Two AP paths versus the stylised cellular comparator.



TWO AP PATHS

3.1

misses per million commands



CELLULAR MODEL · 30 ms HANDOVER

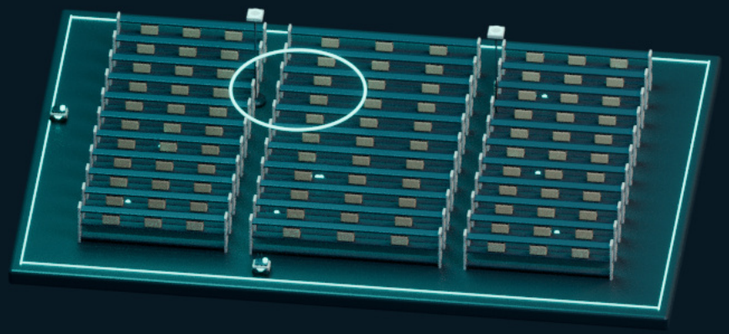
52.8

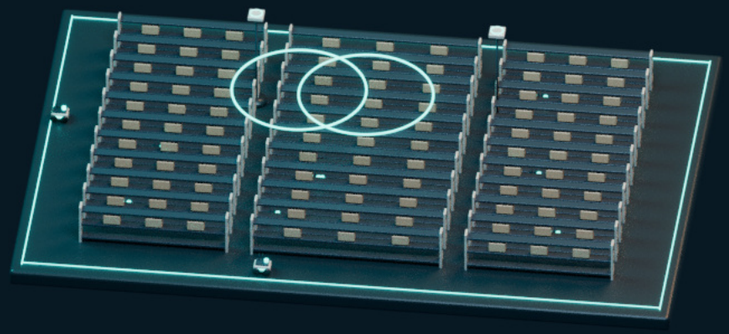
misses per million commands

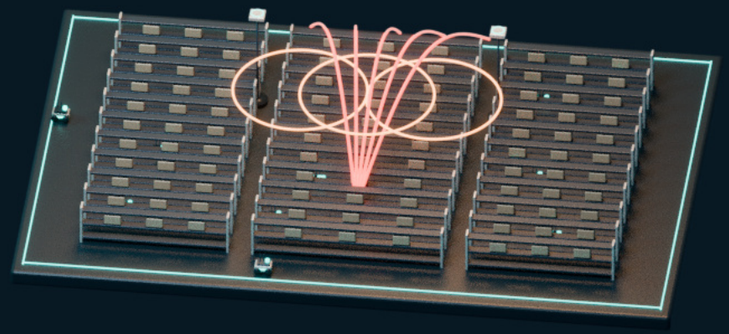
6 paired scenes · 3.6M commands per arm · modelled second client radio

The boundary is visible.

Wi-Fi wins the clean and main cases. The cellular model wins under stress.







CLEAN RADIO

0.6

Wi-Fi misses / million

MAIN CONDITION

3.1

Wi-Fi misses / million

STRESS

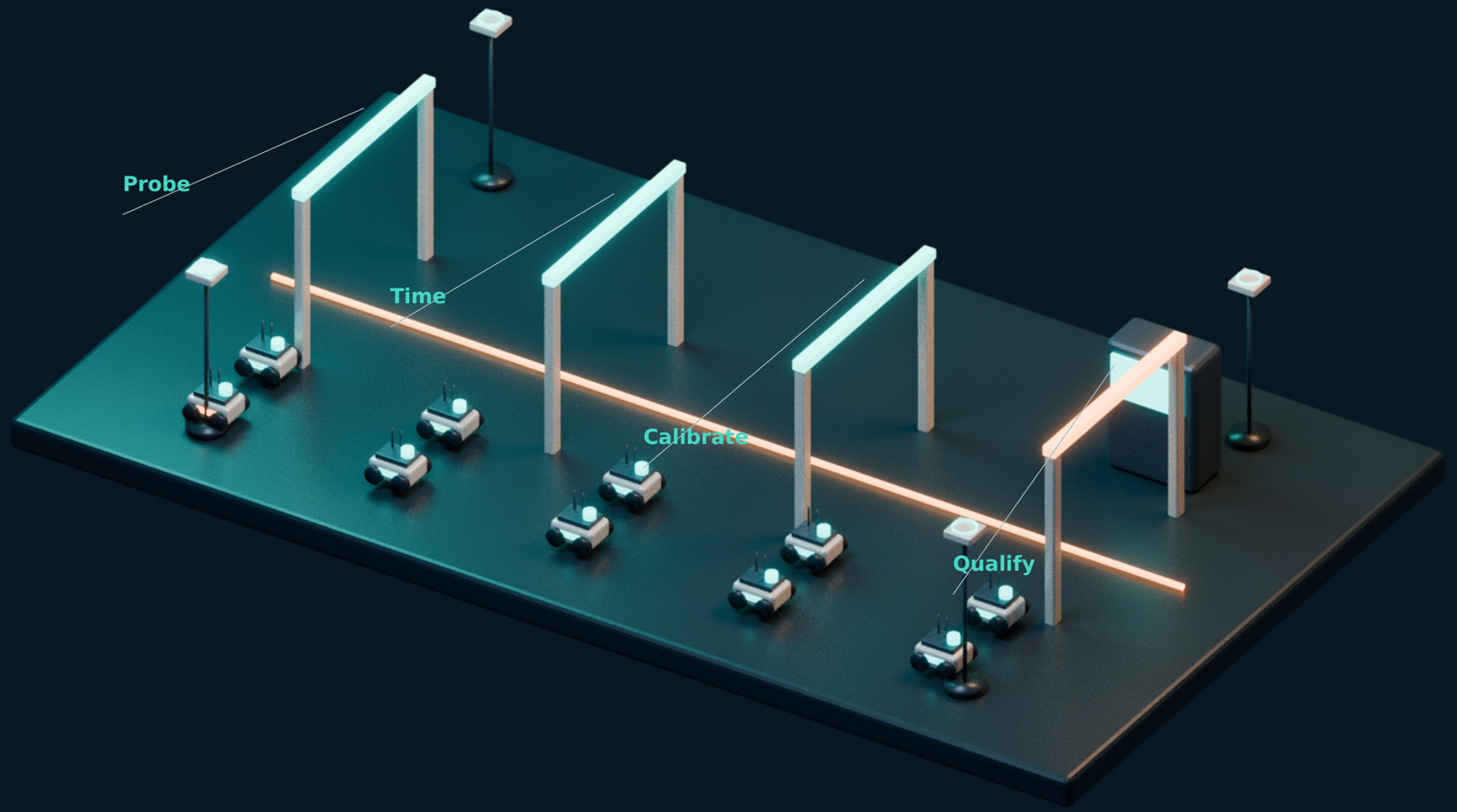
91.4

Wi-Fi misses / million

Cellular model: 52.8 misses / million in all three conditions · Wi-Fi stress reverses the ranking

Measure it. Then scale it.

Probe capability. Time real handovers. Calibrate. Run a controlled fleet trial.



- Working simulator + Studio + tools
- Supervised physical qualification next
- A real private-5G comparison remains pending